

Hypotonia, Infantile, with Psychomotor Retardation and Characteristic Facies 3 (IHPRF3 / TBCK-Related Encephaloneuropathy)

Summary (Answer to the Research Question)

Infantile Hypotonia with Psychomotor Retardation and Characteristic Facies-3 (IHPRF3; OMIM #616900) is an ultrarare **autosomal recessive** neurodevelopmental and **progressive neurodegenerative** disorder caused by **biallelic loss-of-function variants in *TBCK*** (TBC1-domain-containing kinase, chromosome 4q24). Affected children present in infancy with **profound hypotonia, global developmental delay/severe intellectual disability, epilepsy, characteristic (often coarsening) facial dysmorphism, and brain atrophy/white-matter changes**, frequently progressing to a lysosomal-storage-like neurodegenerative course with lipofuscin accumulation (proposed neuronal ceroid lipofuscinosis subtype **CLN15**). Mechanistically, **TBCK loss downregulates mTOR (mTORC1) signaling** and disrupts **RAB5-dependent endolysosomal trafficking and autophagy**; because mTOR remains responsive to exogenous **L-leucine**, leucine supplementation has been proposed as a candidate directed therapy, though care remains largely supportive.

Evidence base: This report synthesizes human clinical case series/cohorts, patient-derived iPSC/neuroprogenitor studies, and neuropathology. No experimental data files were provided; all content is literature-derived.

1. Disease Information

- **Overview:** A severe childhood-onset encephaloneuropathy characterized by infantile hypotonia, psychomotor/developmental delay, intellectual disability, epilepsy, and characteristic facies, with progressive CNS deterioration. Also termed **TBCK deficiency disorder (TBCK-DD)** or **TBCK-related encephaloneuropathy**.

- **Key identifiers:**
- **OMIM:** #616900 (phenotype); *TBCK* gene OMIM 616899.
- **Mondo:** MONDO:0014635 (Hypotonia, infantile, with psychomotor retardation and characteristic facies 3).
- **Orphanet:** ORPHA:494526 (TBCK-related intellectual disability syndrome).
- **ICD-10:** best mapped to G71.2/Q87-range congenital/neuromuscular categories; **ICD-11:** LD24 / 8C (no specific code).
- **MeSH:** no dedicated term; indexed under *Muscle Hypotonia, Intellectual Disability, Neurodevelopmental Disorders*.
- **HGNC:** *TBCK* (HGNC:28348).
- **Synonyms:** IHPRF3; TBCK syndrome; TBCK deficiency disorder; TBCK-related encephaloneuropathy; hypotonia, infantile, with psychomotor retardation and characteristic facies-3; (proposed) neuronal ceroid lipofuscinosis type 15 / CLN15.
- **Data source type:** Aggregated disease-level knowledge derived from **individual patient reports and small cohorts** (WES-diagnosed families) — not EHR-scale datasets. The founding description reported 13 individuals from 9 families

(
P 27040691
 a 2019 review compiled 33 cases
 (

P 30103036
).

2. Etiology

- **Primary cause (genetic):** Biallelic (homozygous or compound heterozygous) pathogenic variants in *TBCK*. "13 individuals from nine unrelated families and affected by likely pathogenic biallelic variants in TBC1-domain-containing kinase (*TBCK*) were identified through whole-exome sequencing"

(
P 27040691
 Autosomal recessive: "an autosomal recessive disorder caused by biallelic pathogenic variants of the *TBCK* gene"
 (

P 36273129
).

- **Genetic risk factors:** The disease is monogenic/Mendelian; the sole causal locus is *TBCK*.
Consanguinity substantially increases risk (multiple homozygous cases in consanguineous/first-cousin unions, e.g.,

P 39865381

).
 No established modifier genes; genotype–phenotype correlation is imperfect
 (

P 32363625

).
- **Environmental risk factors:** None established. Onset is congenital/genetic; no toxin, infectious, or lifestyle trigger is implicated.
- **Protective factors:** None specifically identified. In carriers, a single functional *TBCK* allele is sufficient (parents asymptomatic,

P 36273129

).
- **Gene–environment interactions:** The only therapeutically relevant "environmental" modifier is **dietary L-leucine**, which can reactivate residual mTOR signaling
 (

P 27040691

)
 — an interaction exploited for potential treatment rather than a disease cause.

3. Phenotypes

Core phenotypes (with suggested HPO terms, typical onset, and qualitative frequency):

Phenotype	Type	HPO term	Onset	Frequency
Profound/generalized hypotonia	Clinical sign	HP:0001290 / HP:0001252	Neonatal–infantile	Nearly all (~100%)
Global developmental delay	Sign	HP:0001263	Infantile	~100%
Intellectual disability (often severe/profound)	Sign	HP:0001249 / HP:0010864	Childhood	~100%
Seizures / epilepsy	Sign	HP:0001250	Infancy–childhood	Common (majority)
Facial dysmorphism (coarse, hypertelorism, tented upper lip, exaggerated Cupid's bow, macroglossia, arched eyebrows)	Physical	HP:0000271, HP:0000316, HP:0000219, HP:0000158, HP:0002553	Progressive (coarsening with age)	Variable–common
Brain atrophy / white-matter changes	Lab/imaging	HP:0002059 / HP:0002500	Childhood	Common
Corpus callosum agenesis/hypoplasia	Imaging	HP:0001274	Congenital	Occasional
Respiratory insufficiency / recurrent infections	Sign	HP:0002093	Progressive	Common (major cause of death)
Myopathic changes on biopsy	Lab	HP:0003198	Variable	Reported subset
Cardiac malformations (ToF, aortic arch anomalies)	Sign	HP:0001631 / HP:0001647	Congenital	Rare/emerging
Feeding difficulties / failure to thrive	Sign	HP:0011968 / HP:0001508	Infantile	Common

- **Severity:** Severe to profound; variable across variants. Older children may develop coarse, storage-disorder-like features

(P 27040691).

- **Progression: Progressive/neurodegenerative** — *"pathogenic mutations of the said gene lead to a progressive neurodegenerative disease"*

(
P 30103036
).

- **Quality-of-life impact:** Severe — most affected individuals are non-ambulatory, nonverbal, dependent for all activities of daily living, often requiring gastrostomy feeding and respiratory support.

4. Genetic / Molecular Information

- **Causal gene:** *TBCK* (TBC1-domain-containing kinase), 4q24; HGNC:28348; encodes a protein with **three conserved domains** — **an N-terminal Ser/Thr (pseudo)kinase domain, a central TBC (Rab-GAP) domain, and a C-terminal rhodanese-homology domain (RHOD)** — with long and short isoforms generated by alternative splicing

(
P 33240423
).

Biochemical study of purified full-length protein confirms TBCK is a **catalytically inactive class I pseudokinase** lacking canonical VAIK/HRD/DFG motifs and nucleotide binding

(
P 42546825
).

- **Representative pathogenic variants (ClinVar/literature):**

- c.247C>T, **p.Arg83Ter** (nonsense) — novel in a Chinese patient

(
P 36273129
).

- **p.Gln164*** (nonsense, exon 6) — LOF

(
P 37876076
).

- c.753dup, **p.Lys252*** (frameshift/truncating)

(
P 30103036
).

- **c.1532G>A** (missense) — recurrent; associated with cardiac involvement

(
P 39865381
).

- Early truncating variant with **complete absence of TBCK protein** and myopathy

(
P 32363625
).

- **c.376C>T, p.Arg126* (p.R126X)** — recurrent **founder** stop-gain variant predominating in individuals of **Puerto Rican/"Boricua" ancestry**; defines "TBCK-encephaloneuropathy / Boricua syndrome"

(
P 42563865
).

- **Variant classification:** Pathogenic/likely pathogenic per ACMG/AMP

(
P 36273129
).

Variant types: predominantly nonsense, frameshift, splice, and structural/duplication (LOF); some missense. **Functional consequence: loss of function** — "*TBCK was absent in cells from affected individuals*"

(
P 27040691
).

- **Allele frequency (gnomAD v4, this analysis):** Individual pathogenic alleles are ultrarare. Most pathogenic truncating variants have exome AF <1e-5 (often singletons or absent). The **founder p.Arg126* (c.376C>T; 4-106262103-G-A)** is the most enriched pathogenic LoF allele: **exome AF $\approx 6.0 \times 10^{-5}$ (1/16,569 alleles); genome AF $\approx 2.4 \times 10^{-4}$ (1/4,209 alleles)**, the higher genome frequency reflecting over-representation in admixed/Latino samples. Historical: c.247C>T $\sim 8.2 \times 10^{-6}$ in ExAC

(
P 36273129
).

- **Gene constraint (gnomAD v4):** *TBCK* is **loss-of-function tolerant** — pLI ≈ 0 , oe_lof = 0.87 (90% CI 0.74–1.03; LOEUF ~ 1.03) — consistent with recessive inheritance in which heterozygous carriers are largely unaffected.
- **Allelic architecture:** ClinVar lists **882 TBCK variants**; pathogenic truncating alleles are **distributed across the gene (marked allelic heterogeneity)** with no mutational hotspot other than the Puerto Rican founder allele.
- **Somatic vs germline:** **Germline**, biallelic.
- **Modifier genes:** None validated. Functional partners **PPP1R21** and **FERRY3/C12orf4** form a complex with TBCK

(
P 41789809
)

and are candidate biology-level modifiers.

- **Epigenetics/chromosomal abnormalities:** No specific methylation signature or recurrent chromosomal aberration; chromosomal microarray is typically normal aside from runs of

homozygosity in consanguineous families

(
P 39865381
).

Suggested ontology terms: Gene — *TBCK* (HGNC:28348); MONDO:0014635.

5. Environmental Information

- **Environmental factors:** None causative. **Infectious agents:** none. **Lifestyle factors:** not applicable to disease causation. The only relevant environmental/nutritional variable is **L-leucine** as a candidate therapeutic modulator

(
P 27040691
).

6. Mechanism / Pathophysiology

- **Molecular pathways: mTOR / mTORC1 signaling** — TBCK normally supports mTOR activity; its loss reduces downstream signaling, evidenced by **decreased phospho-S6**: *"decreased phosphorylation of phospho-ribosomal protein S6 was also observed, a finding suggestive of downregulation of mTOR signaling"*

(
P 27040691
).

TBCK also acts as a **RAB5-GAP** within the **TBCK-PPP1R21-FERRY3/C12orf4 complex** governing endo-lysosomal homeostasis

(
P 41789809
).

- **Cellular processes:** Impaired **ER-to-Golgi vesicle transport, autophagosome biogenesis, cell-cycle progression, and cell migration** in patient iNPCs: *"Lack of functional TBCK protein in iNPC is associated with impaired endoplasmic reticulum-to-Golgi vesicle transport and autophagosome biogenesis, as well as altered cell cycle progression and severe impairment in the capacity of migration"*

(
P 35095425
).

Dysregulated **autophagy-lysosome** flux (enhanced mTORC1-mediated autophagosome formation with reduced Rab-mediated autophagosome-lysosome fusion) drives storage

(
P 30591081
).

- **Protein dysfunction:** Loss of function / absent protein; TBCK's Rab-GAP and scaffolding roles are lost.
- **Metabolic/storage changes: Lipofuscin and carbohydrate-rich storage material** accumulate — *"a widespread and massive accumulation of lipofuscin storage material in neurons of the central nervous system"*

(
P 30591081
).

Lysosomal dysfunction impairs mitochondrial quality control, contributing to neurodegeneration

(
P 34816123
).

- **RNA/ribosomal dysregulation:** Multi-omics of TBCK-deficient cells shows **altered ribosomal function, RNA splicing, and miRNA networks**

(
P 41040160
);

patient-derived neurons exhibit **compartment-specific mRNA and lysosomal trafficking defects**

(
P 40093117
),

consistent with a role in RNA transport.

- **Immune involvement:** Not a primary feature; **vacuolated lymphocytes** are a storage marker

(
P 30591081
).

- **Tissue damage mechanism:** Progressive intraneuronal storage and impaired neurogenesis/neuronal migration → neurodevelopmental deficit plus neurodegeneration.
- **Causal chain:** Biallelic *TBCK* LOF → loss of Rab5-GAP activity + reduced mTORC1 (↓ pS6) → defective endolysosomal trafficking, secretory transport, and autophagosome–lysosome fusion → lipofuscin/carbohydrate storage + impaired neuroprogenitor proliferation/migration → hypotonia, developmental delay, and progressive neurodegeneration.
Upstream: *TBCK* loss; **downstream:** mTOR, RAB5/autophagy, storage accumulation, neuronal dysfunction.

Suggested ontology terms: GO:0031929 (TOR signaling); GO:0006914 (autophagy); GO:0016197 (endosomal transport); GO:0005096 (GTPase activator activity); GO:0006888 (ER-to-Golgi vesicle transport); CL:0000031 (neuroblast/neural progenitor), CL:0000540 (neuron), CL:0000127 (astrocyte).

7. Anatomical Structures Affected

- **Organ level: Central nervous system** (primary) — cerebral cortex, white matter, ventricles, corpus callosum; **skeletal muscle** (secondary myopathy); **heart** (rare malformations); systemic storage in **spleen and liver**

(
P 30591081
).

Body systems: nervous, musculoskeletal, occasionally cardiovascular, and respiratory (secondary).

- **Tissue/cell level: Neurons** (lipofuscin storage), **astrocytes** (storage), lymphocytes (vacuolation); neural progenitor cells (migration/proliferation defects).
- **Subcellular level: Lysosome/autophagosome** (GO:0005764, GO:0005776), **endoplasmic reticulum & Golgi** (GO:0005783, GO:0005794), **endosome** (GO:0005768).
- **Localization/lateralization: Bilateral, diffuse** CNS involvement (ventriculomegaly, widened subarachnoid space, atrophy, white-matter changes)

(
P 36273129
).

Suggested UBERON terms: UBERON:0000955 (brain); UBERON:0002316 (white matter); UBERON:0001851 (cortex); UBERON:0002037 (cerebellum); UBERON:0001134 (skeletal muscle tissue).

8. Temporal Development

- **Onset:** Congenital/**infantile** — hypotonia and delay evident in early infancy.
- **Onset pattern:** Insidious/chronic; developmental delay recognized within the first months to ~15 months

(
P 36273129
).

- **Progression: Progressive neurodegenerative** course

(
P 30103036
);

older children develop coarsening facies and storage features

(
P 27040691
).

Course is generally slowly progressive with variable rate; severe forms cause **premature**

death

(
P 30591081
).

- **Duration:** Chronic, lifelong; substantially shortened life expectancy in severe cases.
- **Critical periods:** Early neurodevelopmental window (neurogenesis/migration) is central; any mTOR-directed intervention would plausibly need to be early, though efficacy is unproven.

9. Inheritance and Population

- **Epidemiology:** **Ultrarare**; only tens of cases reported worldwide (33 reviewed by 2019,

P 30103036).

No reliable prevalence/incidence estimates; Orphanet classifies prevalence as <1/1,000,000.

- **Inheritance:** **Autosomal recessive**

(
P 36273129
).

Penetrance: essentially complete in biallelic LOF. **Expressivity:** variable (severity, muscle, cardiac, imaging features). **Heterozygotes:** population-level gnomAD data show LoF tolerance (LOEUF ~1), yet targeted study reports a *mild* neurologic syndrome in some human and mouse heterozygotes

(
P 37353954
).

— i.e., largely recessive with a subtle possible dosage effect. No anticipation (not a repeat-expansion disorder).

- **Consanguinity:** Strong contributor — multiple homozygous cases in consanguineous families

(
P 39865381
).

Founder effect (quantified, gnomAD v4 + 1000 Genomes, this analysis): the p.Arg126* allele (rs575822089) is ~45× **enriched in Admixed American/Latino** samples (AMR genome AF 1.97×10^{-3} , carrier ~1/253) vs Non-Finnish European (AF 4.4×10^{-5}), and reaches **AF $\approx 1.0 \times 10^{-2}$ in 1000 Genomes Puerto Ricans (carrier ~1/49)** — a strong "Boricua syndrome" founder signal

(
P 42563865
).

The allele is essentially absent in African, East Asian, South Asian, Finnish, Ashkenazi

Jewish, and Middle Eastern groups, and **no homozygotes** are seen in gnomAD (consistent with a severe recessive phenotype). **Carrier frequency** for TBCK deficiency overall is low in general (non-founder) populations, consistent with the ultrarare, mostly private pathogenic alleles.

- **Demographics:** Originally reported mainly in individuals of European ancestry; subsequently described across populations including the **first non-Caucasian (Chinese) patient**

(
P 36273129
),

broadening geographic distribution. A pronounced **founder effect in Puerto Rico/Latino (AMR) populations** (p.R126, ~45× *enriched vs NFE*; *PUR carrier ~1/49*) *concentrates cases in the Boricua** population, where prevalence may be higher than expected (4 diagnoses in 6 months at one center;

P 42563865).

Consanguineous populations (e.g., Middle East, South Asia) are also enriched via private homozygous alleles. No strong sex bias (autosomal); age distribution centers on infants/children.

10. Diagnostics

- **Genetic testing (definitive): Whole-exome sequencing** is the primary diagnostic route

(
P 27040691
),

P 39865381);

WGS, multigene hypotonia/neurodevelopmental panels including *TBCK*, or targeted single-gene testing (e.g., for known founder variants) are alternatives. Chromosomal microarray is typically normal (may show runs of homozygosity)

(
P 39865381
).

Variants classified per **ACMG/AMP**

(
P 36273129
).

- **Biochemical/functional:** Absent TBCK protein and **reduced phospho-S6** on Western blot in patient cells (research assay;

(P 27040691).

Vacuolated lymphocytes and tissue storage material on biopsy support a storage phenotype

(P 30591081).

Muscle biopsy may show **nonspecific myopathic changes**

(P 32363625).

- **Imaging:** Brain **MRI** — cerebral/cerebellar atrophy, white-matter changes, ventriculomegaly, widened subarachnoid space, occasional corpus callosum agenesis

(P 27040691),

(P 36273129),

(P 39865381),

and characteristically **periventricular leukomalacia (PVL)-like changes** in the absence of prematurity — a diagnostic clue prompting *TBCK* testing

(P 36522252).

- **RNA-seq diagnostics:** Transcriptome sequencing (fibroblasts > blood) can diagnose *TBCK*-associated encephalopathy missed by conventional ES/CMA

(P 33001864).

Fetal echocardiography/ultrasound may detect cardiac anomalies and cystic hygroma in affected pregnancies

(P 39865381).

- **Clinical criteria / differential diagnosis:** No formal criteria; diagnosis is molecular. **Differentials:** other causes of infantile hypotonia with DD/ID and coarse facies — lysosomal storage disorders (mucopolysaccharidoses, NCLs), congenital myopathies, other *IHPRF* forms (***IHPRF1/NALCN***, ***IHPRF2/UNC80***), Prader-Willi, congenital disorders of glycosylation, and mitochondrial disease. Storage-disease workup is typically negative except for *TBCK*

(P 30591081).

- **Screening:** Not part of standard newborn screening. **Carrier/cascade screening** and **prenatal/preimplantation genetic testing** are informative in known families (utilized for prenatal diagnosis in

(
P 39865381

11. Outcome / Prognosis

- **Survival/mortality:** Guarded; severe forms feature **premature death**

(
P 30591081

commonly from **respiratory failure/recurrent infections** related to hypotonia and neurodegeneration. **Central hypoventilation/respiratory insufficiency is a major, under-recognized complication** — 5/6 in one cohort required respiratory support

(
P 36522252

warranting early respiratory surveillance. No formal survival statistics exist given rarity.

- **Morbidity/function:** Profound disability — most patients are non-ambulatory, nonverbal, and fully dependent, often requiring feeding-tube and respiratory support.
- **Disease course:** Progressive; complications include epilepsy, scoliosis, contractures, aspiration, feeding failure, and infections.
- **Prognostic factors:** Variant severity (complete LOF/absent protein associated with severe multisystem presentations, e.g., myopathy

;
P 32363625

cardiac malformation with c.1532G>A,

).
P 39865381

Neurorehabilitation showed limited benefit in a reported case

(
P 36273129

12. Treatment

- **No approved disease-modifying therapy.** Management is **supportive and multidisciplinary**

(
P 37876076
):

antiseizure medications for epilepsy, physical/occupational/speech therapy, nutritional support (gastrostomy), respiratory care, orthopedic management of scoliosis/contractures.

- **Candidate directed therapy — L-leucine:** Rationale from retained mTOR responsiveness — *"activation of the mTOR pathway in response to L-leucine supplementation was retained, suggesting a possible avenue for directed therapies for this condition"*

(
P 27040691
).

Efficacy remains unproven clinically.

- **Rehabilitation:** Neurorehabilitation training reported with poor response in at least one case

(
P 36273129
).

- **Experimental/future:** mTOR-pathway modulation and endolysosomal/autophagy-targeted approaches are conceptually motivated by the RAB5-GAP/FERRY complex biology

(
P 41789809
);

no registered disease-specific trials established. **Genetic counseling** and reproductive options are central to family management.

Suggested NCIT terms: Supportive Care; Anticonvulsant Therapy; Physical Therapy; Nutritional Support; Leucine (dietary supplement). **CHEBI:** L-leucine (CHEBI:15603); sirolimus/rapamycin (CHEBI:9168) is conceptually contraindicated as an mTOR inhibitor given baseline mTORC1 downregulation.

13. Prevention

- **Primary prevention:** Not applicable (monogenic). **Genetic counseling** for at-risk (especially consanguineous) families; 25% recurrence risk per pregnancy for carrier couples.
- **Reproductive/secondary prevention: Carrier screening, prenatal diagnosis, and preimplantation genetic testing** in known families (demonstrated prenatally in

(
P 39865381
).

Cascade testing of relatives. **Population-targeted carrier screening** for the p.Arg126* founder allele is high-yield in individuals of Puerto Rican/Latino ancestry (PUR carrier

frequency ~1/49; this analysis,

(P 42563865),

and equitable access to genetic testing improves ascertainment

(P 42563865).

- **Tertiary prevention:** Anticipatory management of seizures, aspiration/respiratory infections, feeding difficulties, and orthopedic complications to reduce morbidity.

14. Other Species / Natural Disease

- **Taxonomy/orthologs:** *TBCK* is conserved across vertebrates — human (NCBI Taxon 9606), mouse *Tbck* (NCBI Taxon 10090, Gene ID 271981), zebrafish *tbck* (NCBI Taxon 7955).

Functional conservation demonstrated: mouse *Tbck* loss causes leukoencephalopathy

(P 42182390)

and zebrafish *tbck* is required for heart development

(P 36574072).

- **Natural disease in animals:** No well-characterized naturally occurring animal counterpart is established in OMIA; the disorder is defined in humans and studied via engineered models.

- **Comparative biology:** Evolutionary conservation of the TBC (Rab-GAP), rhodanese, and pseudokinase domains supports cross-species modeling; heterozygous phenotypes are conserved between mice and humans

(P 37353954).

Zoonotic potential: none (non-infectious, genetic).

15. Model Organisms

- **Human cellular models:** Patient-derived iPSCs → induced neuroprogenitor cells (iNPCs) demonstrate impaired ER-to-Golgi transport, autophagosome biogenesis, cell-cycle, and migration

(P 35095425).

Patient-derived neurons show compartment-specific mRNA and lysosomal trafficking defects

([P 40093117](#));

patient fibroblasts/lymphoblasts show absent TBCK, reduced pS6, and autophagic vesicle accumulation

([P 27040691](#)

[P 42563865](#)).

Multi-omics in patient cells reveal ribosomal/RNA-splicing/miRNA dysregulation

([P 41040160](#)).

- **Mouse models (mammalian):** A **novel mouse model of TBCK Syndrome** recapitulates a **degenerative leukoencephalopathy with multisystem involvement**, including neurodevelopmental, **craniofacial**, and **pulmonary** features and a unique enamel/dentin/alveolar-bone **mineralization** defect

([P 42182390](#)

[P 41756984](#)).

Heterozygous (haploinsufficient) *Tbck* variants cause a mild neurologic syndrome in both mice and humans

([P 37353954](#))

— relevant to carrier counseling.

- **Zebrafish (invertebrate/vertebrate screen):** *tbck* knockdown identified it as **essential for heart development (atrioventricular canal formation)**

([P 36574072](#)),

providing a mechanistic basis for the emerging human cardiac phenotype.

- **Phenotype recapitulation:** Mouse models capture CNS leukoencephalopathy, craniofacial, and pulmonary components with good face validity; zebrafish captures cardiac requirement.
- **Limitations:** Cellular models incompletely capture progressive in vivo neurodegeneration; cross-species differences in lifespan/lysosomal storage may limit modeling of the human storage/CLN15 phenotype.

- **Resources:** MGI (mouse *Tbck*), ZFIN (zebrafish *tbck*); NCBI Gene orthologs — mouse *Tbck* (Gene ID 271981), zebrafish *tbck*.

Supported vs. Refuted Hypotheses

Supported: 1. IHPRF3 is autosomal recessive, caused by biallelic LOF in *TBCK*

([P 27040691](#)

[P 36273129](#)

2. Pathophysiology involves mTOR downregulation plus RAB5/endolysosomal-autophagy dysfunction

([P 27040691](#)

[P 35095425](#)

[P 41789809](#)

3. The disorder is progressive/neurodegenerative with lysosomal storage (lipofuscin; proposed CLN15)

([P 30591081](#)

[P 30103036](#)

4. The phenotype extends beyond CNS to muscle and (rarely) heart

([P 32363625](#)

[P 39865381](#)

5. Residual mTOR responsiveness to L-leucine offers a therapeutic rationale

([P 27040691](#)

Refuted / not supported: - Hypotonia is *purely* neurogenic — refuted; primary muscle disease can contribute

([P 32363625](#)

- It is a classical single-substrate lysosomal storage disease — refuted; >20 known LSDs excluded, storage is heterogeneous (P 30591081).

Limitations and Future Directions

- **Rarity:** Only tens of cases; no robust prevalence, survival, or genotype–phenotype statistics.
- **Therapeutics:** Leucine benefit is theoretical; no controlled trials. Need for validated in vivo models and natural-history studies.
- **Mechanism:** Relationship between mTOR downregulation and endolysosomal/RAB5 dysfunction (upstream vs. parallel) requires clarification

(P 35095425)

(P 41789809)

- **Diagnosis:** Broader awareness needed so hypotonia + coarse facies + storage-negative workup prompts *TBCK* sequencing.

Key References (PMID)

27040691 (Bhoj 2016, foundational cohort & mTOR/leucine); 30591081 (Beck-Wödl 2018, CLN15/storage neuropathology); 30103036 (Zapata-Aldana 2019, 33-case review, progressive neurodegeneration); 35095425 (Moreira 2021, iNPC secretory/autophagy defects); 32363625 (Saredi 2020, myopathy); 36273129 (Tan 2022, first non-Caucasian case, ACMG); 37876076 (Chand 2023, LOF nonsense case); 39865381 (Mastromoro 2025, cardiac involvement, prenatal); 41789809 (Chen 2026, TBCK–PPP1R21–FERRY3 RAB5-GAP complex); 42563865 (De Luca-Ramirez 2023, Puerto Rican p.R126*/Boricua founder); 36522252 (Sabanathan 2023, hypoventilation & PVL-like MRI); 33240423 (Wu 2021, three-domain structure/isoforms); 42546825 (Maurya 2026, class I pseudokinase); 34816123 (Tintos-Hernández 2021, lysosomal-mitochondrial quality control); 41040160 (Diaz-Rosado 2025, ribosome/RNA-splicing/miRNA multi-omics); 40093117 (Flores-Mendez 2025, mRNA & lysosomal trafficking in patient neurons); 37353954 (Nair 2023, heterozygote mild syndrome, mouse); 42182390 (Melendez-Perez 2026, mouse model); 41756984 (Katsura 2026, mouse mineralization/leukoencephalopathy); 36574072 (Ma 2022, zebrafish tbck heart development); 33001864 (Murdock 2021, RNA-seq diagnostics).

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery